

Air Pollution Dashboard Technical Report

1 Introduction

This report aims to detail the data analysis lifecycle for our air pollution dashboard. The dashboard designed for Finnish residents, enabling them to view air pollution levels across northern Europe, filter by key air pollutants, and obtain forecasts for different locations. Through interactive visualizations, Finnish residents can make informed decisions about engaging in outdoor activities or traveling to various parts of Europe. The pollutants we planned to analyse are selected based on recommendations by the 2008/50/EC EU directive [1]. Their impacts are as follows:

- Sulphur dioxide [2], a byproduct of the combustion of sulphur-containing fuels, is a component of acid rain which raises the acidity level of soils, hurting agriculture, and land and aquatic systems.
- Excessive atmospheric nitrous oxides [3] lead to eutrophication.
- Lead [4] results in neurological development issues in children.
- Carbon monoxide [5], a colourless and odourless gas, diminishes the human body's ability to absorb oxygen and in high concentrations, results in death.
- Benzene [6] is a carcinogenic compound released from petroleum product and cigarette smoke.
- PM10 and PM2.5 [5] are airborne particles with diameters of 10 and 2.5 μm respectively. Their small size allows them to be inhaled and accumulate in internal organs.
- Ground level ozone [7], a formed by reactions nitrogen oxides and volatile organic compounds, can inflame the respiratory system and oxidise other pollutants in the air to create smog.

We initially aimed to visualize data for all European countries, but the model size and sparse data in some regions made this impractical. Therefore, we limited the scope to Northern Europe.

2 Data Collection

Two datasets were used for visualization model training: the air pollution dataset and the metadata.

2.1 Air pollution concentrations

The air pollution data was sourced from the European Environment Agency's Air Quality Download Service [8], which provides pollutant data by country, city, pollutant type, and date range. To avoid large parquet files and complex HTTP requests, we downloaded a CSV file containing URLs, each linking to data from a specific monitoring station and pollutant. This method enabled easy sharing without exceeding GitHub's storage limits. Two datasets were used: up to date but unverified data (E2a, from 2025) and verified data (E1a, 2023–2025). Since E2a alone was insufficient for model training, both datasets were combined.

2.2 Metadata

The metadata contains information about sampling locations including but not limited to station code, GPS coordinates, the type of pollutant it measures, and even the height of the sensor from the ground.

3 Data Preprocessing

The data preprocessing step involves recoding data types, selecting columns, imputing invalid values, removing duplicates, and checking for outliers. Both air pollution datasets are then joined to map each air pollution concentration value to its GPS coordinates.

3.1 Preprocessing of air pollution data

The procedure for processing air pollution data is as follows:

1. Filter values based on the date range.
2. Delete records marked as invalid, which accounts for ~11% of the data. Invalid data can show up for 24 hours successively, which does not make sense to impute.
3. Convert measurement timestamps into datetime format, and pollutant concentrations to float.

- Mark negative measurements as null values and impute using forward fill. The magnitudes of negative measurements are negligible ($\sim 4 \text{ ug/m}^3$) and are attributed to sensor noise with preceding measurements close to 0 ug/m^3 .

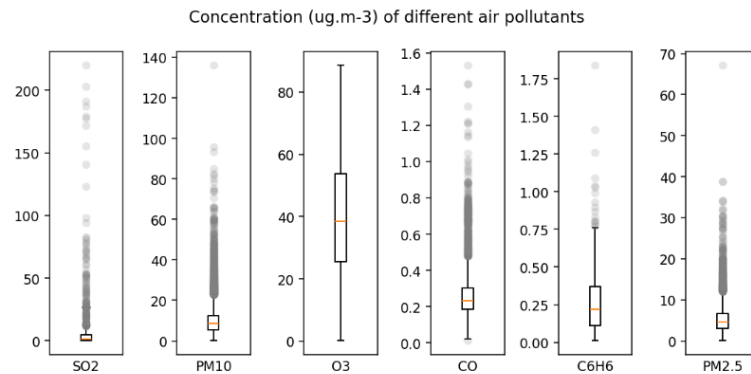


Figure 1 Distribution of pollutant concentrations

Pollutant	Reference time	Good	Fair	Moderate	Poor	Very poor	Extremely poor
PM _{2.5}	1h	0-5	6-15	16-50	51-90	91-140	>140
PM ₁₀	1h	0-15	16-45	46-120	121-195	196-270	>270
O ₃	1h	0-60	61-100	101-120	121-160	161-180	>180
NO ₂	1h	0-10	11-25	26-60	61-100	101-150	>150
SO ₂	1h	0-20	21-40	41-125	126-190	191-275	>275

Figure 2 Pollutant concentrations and health indicators

The resulting data is examined to ascertain if pollutant concentrations are reasonable. In Figure 1, the outliers of the boxplots for each pollutant fall within the ranges of the health indicators [9] in Figure 2, supporting the veracity of the processed data.

3.2 Preprocessing of metadata

Columns and duplicate records were dropped from the metadata dataset, leaving the columns: Country, Air quality station name, Sampling point ID, Air pollutant, Longitude and Latitude.

4 Exploratory Data Analysis

We analysed summary statistics and explored differences in space, time, and pollutant types.

```

Num of records BEFORE filtering: 54789
Num of records AFTER filtering: 48793
Pollutant
PM10      17748
PM2.5     14033
O3         7336
SO2        6151
CO         3187
C6H6       338
Name: count, dtype: int64

```

Table 1 Count of records for air pollution data

	Pollutant	Start	Value
count	48793.000000	48793	48793.000000
mean	1729.692907	2025-10-04 11:27:10.783104256	11.532878
min	1.000000	2025-10-01 00:00:00	0.000001
25%	5.000000	2025-10-02 17:00:00	2.805905
50%	7.000000	2025-10-04 11:00:00	6.370300
75%	6001.000000	2025-10-06 06:00:00	12.100000
max	6001.000000	2025-10-08 00:00:00	219.915546
std	2713.943457	NaN	15.078044

Table 2 Summary statistics of air pollution data

Table 1 shows large discrepancies in record counts across pollutants, as not all stations measured every air pollutant (Figure 3) or sampled hourly. Therefore, our dashboard focused on a smaller selection of pollutants: PM_{2.5}, PM₁₀, sulphur dioxide, ground level ozone, carbon monoxide, and benzene. Lead and nitrous oxides were omitted from the list of pollutants of interest.

Concentration (ug.m-3) of air pollutants at different stations

Legend:

- O3
- PM10
- PM2.5
- SO2

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3

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***** Percentile: 1% *****
Pollutant: SO2
Mean: 28.53, Std: 992.98, Min: 0.01, Q1: 0.20, Median: 0.43, Q3: 0.76, Max: 35836.99
Pollutant: PM10
Mean: 28.36, Std: 29.67, Min: 0.30, Q1: 7.88, Median: 14.87, Q3: 41.73, Max: 195.23
Pollutant: O3
Mean: 90.53, Std: nan, Min: 90.53, Q1: 90.53, Median: 90.53, Q3: 90.53, Max: 90.53
Pollutant: PM2.5
Mean: 110.24, Std: 990.18, Min: 0.10, Q1: 6.25, Median: 8.80, Q3: 12.60, Max: 9985.00

***** Percentile: 10% *****
Pollutant: SO2
Mean: 26.89, Std: 944.93, Min: 0.01, Q1: 0.21, Median: 0.48, Q3: 0.87, Max: 35836.99
Pollutant: PM10
Mean: 23.27, Std: 37.76, Min: 0.01, Q1: 8.44, Median: 14.41, Q3: 24.55, Max: 4490.96
Pollutant: O3
Mean: 84.19, Std: 5.84, Min: 77.97, Q1: 80.78, Median: 81.33, Q3: 90.35, Max: 90.53
Pollutant: PM2.5
Mean: 29.93, Std: 427.62, Min: 0.00, Q1: 5.48, Median: 8.90, Q3: 13.43, Max: 9985.00

***** Percentile: 30% *****
Pollutant: SO2
Mean: 10.72, Std: 355.85, Min: 0.00, Q1: 2.10, Median: 6.30, Q3: 9.31, Max: 35836.99
Pollutant: PM10
Mean: 18.75, Std: 26.56, Min: 0.01, Q1: 7.57, Median: 12.52, Q3: 20.57, Max: 4490.96
Pollutant: O3
Mean: 65.93, Std: 19.24, Min: 0.00, Q1: 54.49, Median: 66.97, Q3: 78.63, Max: 145.73
Pollutant: PM2.5
Mean: 13.71, Std: 215.98, Min: 0.00, Q1: 3.97, Median: 6.61, Q3: 11.00, Max: 9985.00

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Figure 6 Analysis of four pollutants of the most polluted cities

To assess the pollution severity across different scales, we analysed the top 1%, 10%, and 30% most polluted cities, capturing both extreme hotspots and broader high-pollution areas in Figure 6. Some extreme values may reflect sampling errors rather than true pollution levels. In the top 1% of the cities, PM2.5 shows the highest mean concentration and a large variation. Even excluding sampling errors, the top 1% areas still maintain significantly high PM2.5 levels. Overall, severe pollution is concentrated in a few areas, particularly for PM2.5, with a substantial gap between the top 1% and top 10% most polluted areas.

5 Data Visualization

To visualise time and space simultaneously, an interactive map with an hourly slider is used.

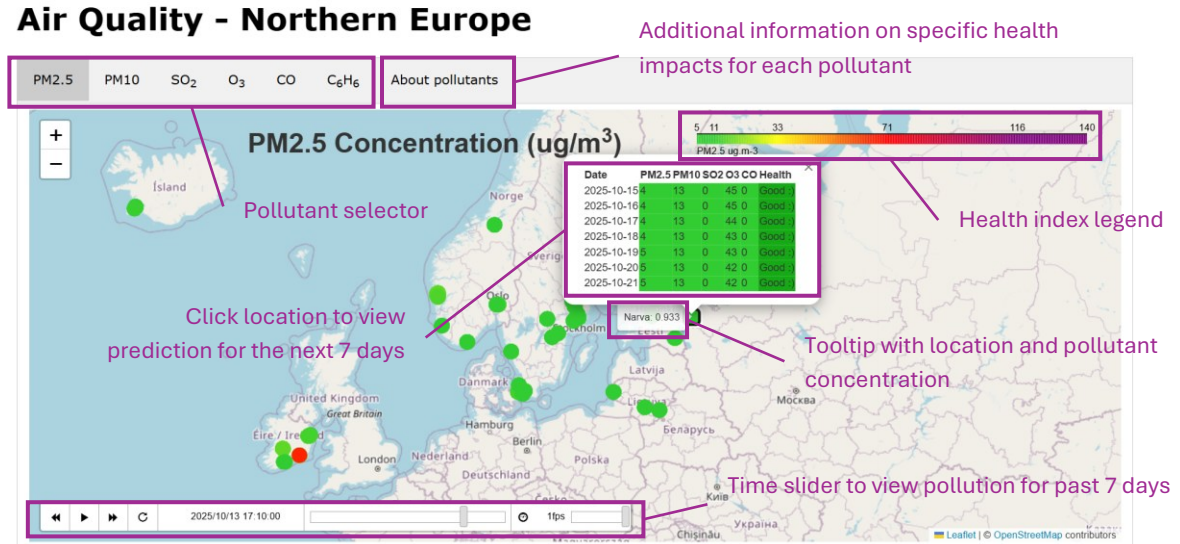


Figure 7 Dashboard interactive elements

There is one map for each pollutant. Users can select the hour and which pollutant to view, as well as manipulate the map by zooming and dragging.

6 Learning and Predictions

Initially, this project aimed to use linear regression, but it proved ill-suited for time-series data due to its inability to capture temporal dependencies. Consequently, Gated Recurrent Units (GRUs) were adopted as they are specifically designed to model sequential data and long-term dependencies.

To enhance performance, the model incorporated moving averages as input features to smooth noise and provide a more stable representation of historical data. It also included static station and country information as static features to account for geographical and regional variations in pollution levels.

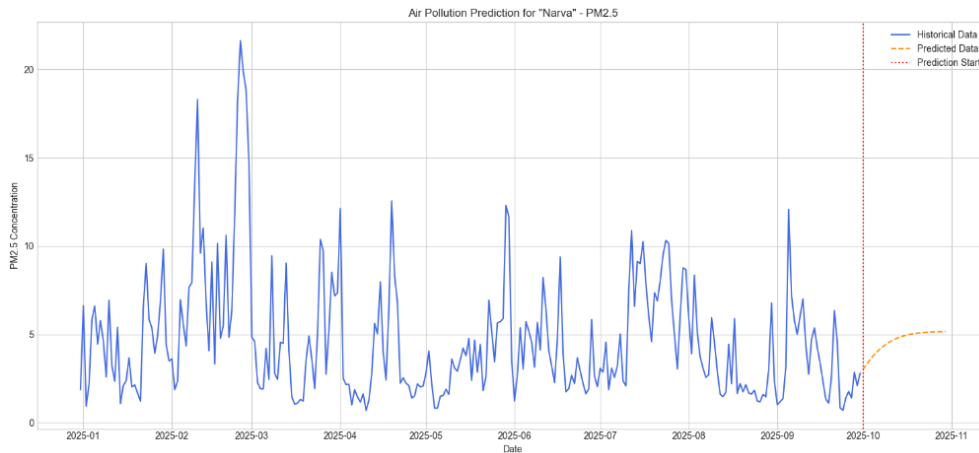


Figure 8 The historical data and predictions for station “Narva” on PM2.5

A key challenge emerged with the initial plan to predict hourly pollution levels for the next year. While the model performed adequately on Finland's dataset, scaling this approach to include data from all countries resulted in prohibitively long training times. This was due to the sheer volume of data and the computational complexity of the GRU models. To address this computational bottleneck and balance complexity with feasibility, the prediction scope was strategically adjusted. The horizon was reduced to one month, and the granularity was shifted from hourly to daily. This change drastically reduced the dataset size and computational requirements, making the pipeline more efficient. The pipeline's modular design, which trains separate models for each pollutant, also worked well, allowing for pollutant-specific optimization and better overall accuracy. After training, this model generates a 30-day forecast. Figure 8 shows the historical data and predictions for station “Narva” on PM2.5 together.

Looking ahead, the pipeline could be enhanced to support rolling updates, allowing the model to incorporate new daily data and improve its predictions dynamically. This would make the system more adaptive and suitable for real-time applications. Exploring ensemble methods that combine GRUs with other models could also improve prediction accuracy. Implementing early stopping and hyperparameter tuning earlier in the process could have optimized the training process and reduced overfitting. Finally, integrating external factors such as weather data or traffic patterns could provide a more comprehensive understanding of pollution dynamics.

7 Communication of Results

The visualizations and predictions are presented on a website, making it accessible worldwide. To allow users to decide if it is safe for them to travel to other northern European countries or outdoors, sampling points are coloured based on the concentration of pollutants. The health index legend (Figure 7, item 1) maps the pollutant concentration to health impacts based on EU health indicators in Figure 2. Tooltips reveal the location name of the sampling point and the exact pollutant concentration. Users can click on the coloured dots on the map to view that location's air quality forecast for the next 7 days. Curious users can switch between tabs to view maps for different pollutants.

The website is hosted on GitHub Pages as a static HTML page. The interactive maps are rendered using the TimeStampedGeoJSON plugin from the python Folium library. GitHub Actions could be added in the future to automate periodic data updates, incorporating API requests from the European Environment Agency's Air Quality Download Service. Since weather conditions and wind direction affect the distribution of air pollutants, these factors could be incorporated into the prediction model and visualization in future iterations.

8 References

Due to page limit restrictions, please find the references on our GitHub repository:

<https://ngdeqi.github.io/DATA11001-Project/Technical-Report-References.pdf>